MTH 202 Chapter 1: Three-Dimensional Space & Vectors

1.4 Cross Products

Motivation: We want an operation that will take in two vectors and spit out a vector orthogonal to both vectors.

> **Definition:** If $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$, then the cross product of **a** and **b** is the vector:
> $\mathbf{a} \times \mathbf{b} = \langle a_2 b_3 - a_3 b_2, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1 \rangle$.

Remark: The calculation of the cross product is closely related to finding the determinant of a 3 by 3 matrix. So much so, that we often write:

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2 b_3 - a_3 b_2)\mathbf{i} - (a_1 b_3 - a_3 b_1)\mathbf{j} + (a_1 b_2 - a_2 b_1)\mathbf{k} = \langle a_2 b_3 - a_3 b_2, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1 \rangle.$$

Note: Unlike the dot product the result from the cross product is a vector!
Note: While the dot product can be defined in any number of dimensions the cross product is only defined for three dimensional vectors!

Example 1: If $\mathbf{a} = \langle 2, 3, -3 \rangle$ and $\mathbf{b} = \langle 1, -1, 2 \rangle$ then determine each of the following cross products $\mathbf{a} \times \mathbf{b}$ for each of the following:

a) $\mathbf{a} \times \mathbf{b}$

We calculate that: $\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 3 & -3 \\ 1 & -1 & 2 \end{vmatrix} = \langle 6 - 3, -(4 - (-3)), -2 - 3 \rangle = \langle 3, -7, -5 \rangle$.

b) $\mathbf{b} \times \mathbf{a}$.

We calculate that: $\mathbf{b} \times \mathbf{a} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -1 & 2 \\ 2 & 3 & -3 \end{vmatrix} = \langle 3 - 6, -(-3 - 4), 3 - (-2) \rangle = \langle -3, 7, 5 \rangle$.

Note: $\mathbf{a} \times \mathbf{b} \neq \mathbf{b} \times \mathbf{a}$, but $\mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a})$

c) $\mathbf{a} \times \mathbf{a}$.

We calculate that: $\mathbf{a} \times \mathbf{a} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 3 & -3 \\ 2 & 3 & -3 \end{vmatrix} = \langle -9 - (-9), -(-6 - (-6)), 6 - 6 \rangle = \langle 0, 0, 0 \rangle = \vec{0} = \mathbf{0}$.

*Note: Using the dot product, one can observe that the results in (a) & (b) for this example are orthogonal to both vector **a** and vector **b**.*

Theorem: Let $\boldsymbol{a}$ and $\boldsymbol{b}$ be vectors in 3-space. Then, $\boldsymbol{a}\times\boldsymbol{b}$ is orthogonal to both $\boldsymbol{a}$ and $\boldsymbol{b}$.

Proof (Not done in class):
We know if $\mathbf{a}=\langle a_1,a_2,a_3\rangle$ and $\mathbf{b}=\langle b_1,b_2,b_3\rangle$, then $\mathbf{a}\times\mathbf{b}=\langle a_2b_3-a_3b_2, a_3b_1-a_1b_3, a_1b_2-a_2b_1\rangle$. So we calculate that:

$$\begin{aligned}&\mathbf{a}\cdot(\mathbf{a}\times\mathbf{b})=a_1(a_2b_3-a_3b_2)+a_2(a_3b_1-a_1b_3)+a_3(a_1b_2-a_2b_1)\\&=a_1a_2b_3-a_1a_3b_2+a_2a_3b_1-a_2a_1b_3+a_3a_1b_2-a_3a_2b_1=0.\end{aligned}$$

We also calculate that

$$\begin{aligned}&\mathbf{b}\cdot(\mathbf{a}\times\mathbf{b})=b_1(a_2b_3-a_3b_2)+b_2(a_3b_1-a_1b_3)+b_3(a_1b_2-a_2b_1)\\&=b_1a_2b_3-b_1a_3b_2+b_2a_3b_1-b_2a_1b_3+b_3a_1b_2-b_3a_2b_1=0.\end{aligned}$$

So, we have the desired result by the definition of orthogonal.

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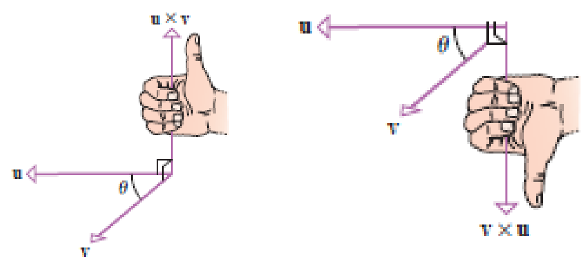
A question that we might have is regarding the direction in which $\mathbf{a}\times\mathbf{b}$ points if **a** and **b** are two non-parallel vectors. While this is possible to determine by investigating the resulting vector $\mathbf{a}\times\mathbf{b}=\langle a_2b_3-a_3b_2, a_3b_1-a_1b_3, a_1b_2-a_2b_1\rangle$, the direction can be summarized into what we call the ***right-hand rule***.



It is important to note that we do not *prove* this rule here.

Remark: The following identities may remind you of the right hand rule from physics (see page 799 of the book for more details): $\mathbf{i}\times\mathbf{j}=\mathbf{k}$, $\mathbf{j}\times\mathbf{i}=-\mathbf{k}$, $\mathbf{j}\times\mathbf{k}=\mathbf{i}$, $\mathbf{k}\times\mathbf{j}=-\mathbf{i}$, $\mathbf{k}\times\mathbf{i}=\mathbf{j}$, and $\mathbf{i}\times\mathbf{k}=-\mathbf{j}$.

Theorem: If θ is the angle between **a** and **b**, then $\|\mathbf{a}\times\mathbf{b}\|=\|\mathbf{a}\|\|\mathbf{b}\|\sin(\theta)$.

Proof: See page 799 of the text.

Corollary: Two nonzero vectors **a** and **b** (in three dimensions) are parallel if and only if $\mathbf{a}\times\mathbf{b}=\mathbf{0}$.

Proof: We know that two vectors in three dimensions are parallel if and only if $\theta=0$ *or* π (think about why this is so). In either case $\|\mathbf{a}\times\mathbf{b}\|=\|\mathbf{a}\|\|\mathbf{b}\|(0)=0$, and $\mathbf{a}\times\mathbf{b}=\mathbf{0}$.

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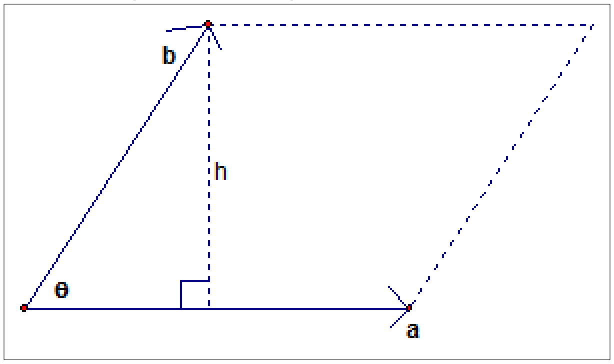
Example 2: Suppose that **a** and **b** are vectors in three dimensions. Find a simple formula for the area of the parallelogram determined by **a** and **b**.

A picture of the situation (assuming that the angle is not obtuse) is:



We know that the area of the parallelogram is $A = h\|\mathbf{a}\|$. Simple right triangle trigonometry tells us that $h = \|\mathbf{b}\|\sin(\theta)$. So, by what we know from this section: $A = \|\mathbf{b}\|\|\mathbf{a}\|\sin(\theta) = \|\mathbf{a}\times\mathbf{b}\|$.

In the case where the angle is obtuse we obtain $h = \|\mathbf{b}\|\sin(\pi-\theta) = \|\mathbf{b}\|\sin(\theta)$ and we get the same formula.

Theorem: The area, A, of a parallelogram that has **a** and **b** as adjacent sides is given by $A = \|\mathbf{a}\times\mathbf{b}\|$.

*Question: How would the formula change if you were asked to find the area of the triangle determined by **a** and **b**?*

Theorem: Suppose **a**, **b**, and **c** are three dimensional vectors and c is a scalar. Then,

1. $\mathbf{a}\times\mathbf{b} = -\mathbf{b}\times\mathbf{a}$
2. $(c\mathbf{a})\times\mathbf{b} = c(\mathbf{a}\times\mathbf{b}) = \mathbf{a}\times(c\mathbf{b})$
3. $\mathbf{a}\times(\mathbf{b}+\mathbf{c}) = \mathbf{a}\times\mathbf{b}+\mathbf{a}\times\mathbf{c}$
4. $(\mathbf{a}+\mathbf{b})\times\mathbf{c} = \mathbf{a}\times\mathbf{c}+\mathbf{b}\times\mathbf{c}$
5. $\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}) = (\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}$
6. $\mathbf{a}\times(\mathbf{b}\times\mathbf{c}) = (\mathbf{a}\cdot\mathbf{c})\mathbf{b}-(\mathbf{a}\cdot\mathbf{b})\mathbf{c}$

Proof: These properties can be proven with typical component wise proofs.

Definition: If $\mathbf{u} = \langle u_1, u_2, u_3\rangle$, $\mathbf{v} = \langle v_1, v_2, v_3\rangle$, & $\mathbf{w} = \langle w_1, w_2, w_3\rangle$ are vectors in 3-space, then the number

$$\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})$$

is called the <u>scalar triple product of **u**, **v**, and **w**</u>.

Note: There is no ambiguity to the order of vector operations if this expressed as $\mathbf{u}\cdot\mathbf{v}\times\mathbf{w}$.

Example 3: Let $\mathbf{u}=2\mathbf{i}-\mathbf{j}+\mathbf{k}$, $\mathbf{v}=\mathbf{i}-2\mathbf{j}-\mathbf{k}$, & $\mathbf{w}=3\mathbf{i}+2\mathbf{j}-2\mathbf{k}$. Compute the scalar triple product.

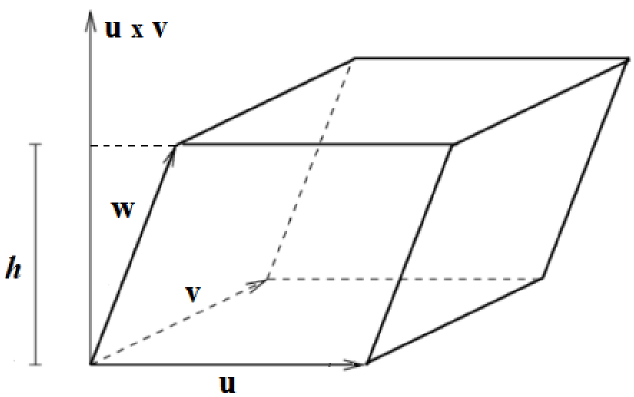
We note that $\mathbf{v}\times\mathbf{w}=\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ 1 & -2 & -1\\ 3 & 2 & -2\end{vmatrix}=\langle 4-(-2),-(-2-(-3)),2-(-6)\rangle=\langle 6,-1,8\rangle$.

Therefore, $\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})=(2\cdot 6)+(-1\cdot -1)+(1\cdot 8)=12+1+8=21$

Definition: A <u>parallelepiped</u> is a solid that has 6 quadrilateral faces such that opposing faces are congruent parallelograms.

Example 4: Suppose that you are working with three non-zero vectors, **u**, **v**, and **w** which form adjacent sides of a parallelepiped.

a) What is the volume of the resulting parallelepiped?



We can find the volume of the figure by determining the area of the base and multiplying by the height of the parallelepiped. We see that:

Area of base = $\|\mathbf{u}\times\mathbf{v}\|$

Height = $comp_{\mathbf{u}\times\mathbf{v}}(\mathbf{w})=\dfrac{(\mathbf{u}\times\mathbf{v})\cdot\mathbf{w}}{\|\mathbf{u}\times\mathbf{v}\|}$

Therefore the total volume of the parallelepiped is:

$$V=base\cdot height=\|\mathbf{u}\times\mathbf{v}\|\left(\frac{(\mathbf{u}\times\mathbf{v})\cdot\mathbf{w}}{\|\mathbf{u}\times\mathbf{v}\|}\right)=(\mathbf{u}\times\mathbf{v})\cdot\mathbf{w}$$

This is the triple scalar product!!!

b) Determine the volume of the parallelepiped determined by the vectors in Example 3.

Immediately we observe that the volume of the parallelepiped must be 21!

Theorem: Let **u**, **v**, and **w** be nonzero vectors in 3-space.

a) The volume V of the parallelepiped that has **u**, **v**, and **w** as adjacent edges is given by

$$V = \left|\mathbf{u}\cdot\left(\mathbf{v}\times\mathbf{w}\right)\right|$$

b) $\mathbf{u}\cdot\left(\mathbf{v}\times\mathbf{w}\right)=0$ if and only if **u**, **v**, and **w** lie in the same plane.

Proof (Not done in class):

We note that a) is proved above in Example 4a).

To prove b) we just need to note that if **u**, **v**, and **w** lie in the same plane then the height of the parallelepiped is 0 and therefore has volume 0 which means that $\mathbf{u}\cdot\left(\mathbf{v}\times\mathbf{w}\right)=0$. Similarly if $\mathbf{u}\cdot\left(\mathbf{v}\times\mathbf{w}\right)=0$ then the parallelepiped has volume 0 and thus the vectors **u**, **v**, and **w** must lie in the same plane.

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Example 5: Do the vectors from Example 3 all reside within the same plane?

Clearly not, since $\mathbf{u}\cdot\left(\mathbf{v}\times\mathbf{w}\right)\neq 0$. As stated in the above theorem, the only way for 3 vectors to be co-planar is for $\mathbf{u}\cdot\left(\mathbf{v}\times\mathbf{w}\right)=0$.